

QUICK REFERENCE CARD

Print this and keep it on your desk! 



CRITICAL DATES

Date	Event	Action
Oct 30	MIDTERM	Submit proposal + present (5 min)
Dec 4	FINAL	Submit project + present (10 min)



GRADING AT A GLANCE

Midterm (25% of grade):

- Problem & Solution: 20 pts
- Technical Approach: 25 pts
- Data Plan: 15 pts
- Implementation Plan: 15 pts
- Presentation: 15 pts
- GitHub Setup: 10 pts **Total: 100 pts**

Final (35% of grade):

- System Works: 35 pts (most important!)
- Code Quality: 20 pts
- Results: 15 pts
- GitHub Repository: 10 pts
- Demo Video: 10 pts
- Presentation: 10 pts **Total: 100 pts**

Bonus Points:

- Tier 2: +10 pts
- Tier 3: +20 pts
- Other bonuses: up to +15 pts



PROJECT TIERS

Tier 1 (  ☆☆☆)

- Simple CV project
- Pre-trained models
- Colab free tier
- 100 points possible

Tier 2 (★ ★ ★ ☆ ☆)

- Multi-component OR VLM
- Some fine-tuning
- May need Heidi/\$5-10
- 110 points possible

Tier 3 (★ ★ ★ ★ ★)

- Production-quality
- Novel approach
- Significant effort
- 120 points possible



WHAT TO SUBMIT

Midterm (Oct 30):

- ☐ Presentation slides (PDF/PPTX)
- ☐ GitHub repository
- ☐ README.md (complete)
- ☐ 5-minute presentation

Final (Dec 4):

- ☐ Complete GitHub repository
- ☐ Demo video (3-5 min)
- ☐ 10-minute presentation (with live demo!)
- ☐ AI usage log



FREE RESOURCES

GPU:

- Google Colab: ~12 hrs/day
- Kaggle: 30 hrs/week
- Heidi: Ask instructor

Models:

- YOLOv8: Object detection
- ResNet: Classification
- BLIP: Image captioning
- All free from Hugging Face!

Datasets:

- Roboflow Universe
- Kaggle Datasets
- Hugging Face Datasets
- COCO, ImageNet



WEEKLY GOALS

Week 9: Choose project + dataset **Week 10:** Submit proposal ⚠️ **Week 11:** Get model working **Week 12:** Improve performance **Week 13:** Create demo video **Week 14:** Final polish **Week 15:** Present! 🎉



AI USAGE RULES



ALLOWED:

- Learn concepts
- Debug code
- Generate non-CV code (HTML, etc.)
- Optimize code



NOT ALLOWED:

- Build entire project
- Copy without understanding
- Skip learning



REQUIRED:

- Document in AI usage log
- 5-10 entries minimum
- Show what you learned



EMERGENCY CONTACTS

Stuck?

1. Check FAQ document

2. Canvas discussion board
3. Office hours
4. Email instructor

Don't wait until it's too late!



DAILY CHECKLIST

- ☐ Work on project (30 min min)
 - ☐ Commit to GitHub
 - ☐ Document AI usage
 - ☐ Test code
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SUCCESS FORMULA

1. **Start early** (Week 9!)
 2. **Choose Tier 1** (if unsure)
 3. **Use free resources**
 4. **Document everything**
 5. **Ask for help early**
 6. **Test frequently**
 7. **Follow weekly checklist**
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TOP 5 MISTAKES TO AVOID

1. **✗** Starting too late
 2. **✗** Choosing too ambitious project
 3. **✗** Not testing code frequently
 4. **✗** Not documenting AI usage
 5. **✗** Not asking for help
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COMMON CODE SNIPPETS

Test GPU:

```
!nvidia-smi
```

Install packages:

```
!pip install ultralytics torch opencv-python
```

Load YOLO:

```
from ultralytics import YOLO
model = YOLO('yolov8n.pt')
```

Run detection:

```
results = model('image.jpg')
results[0].show()
```

REPOSITORY STRUCTURE

```
project-name/
├─ README.md
├─ requirements.txt
├─ data/
├─ models/
├─ notebooks/
│   └─ 04_demo.ipynb
├─ results/
└─ docs/
    └─ AI_usage_log.md
```

PRESENTATION CHECKLIST

Midterm (5 min):

- ☐ Problem & solution
- ☐ Technical approach
- ☐ Data plan
- ☐ Timeline

Final (10 min):

- ☐ Problem & solution (2 min)
- ☐ LIVE DEMO (5 min) 
- ☐ Results (2 min)
- ☐ Learnings (1 min)

QUICK TROUBLESHOOTING

"CUDA out of memory" → Reduce batch size

"Can't find dataset" → Check file paths

"Model won't load" → Verify .pt file exists

"Training is slow" → Check GPU is enabled

"Code won't run" → Check requirements.txt



PERFORMANCE TARGETS

Object Detection:

- Okay: mAP50 > 0.5
- Good: mAP50 > 0.7
- Great: mAP50 > 0.9

Classification:

- Okay: Accuracy > 70%
- Good: Accuracy > 85%
- Great: Accuracy > 95%

Speed:

- Real-time: >20 FPS
 - Interactive: >5 FPS
 - Batch: <1 sec/image
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REMEMBER

Build > Write Focus on creating a working system, not writing essays!

Progress > Perfection 70% working is better than 100% not working!

Learn > Copy Use AI to learn, not to avoid learning!

Ask > Struggle Questions are good! Suffering in silence is bad!



IMPORTANT LINKS

- Colab: colab.research.google.com
 - GitHub: github.com
 - Roboflow: universe.roboflow.com
 - YOLO Docs: docs.ultralytics.com
 - PyTorch: pytorch.org
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FINAL TIP

This card is your survival guide. Check it daily!

Print it. Pin it. Use it. Succeed! 

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Start coding or generate with AI.